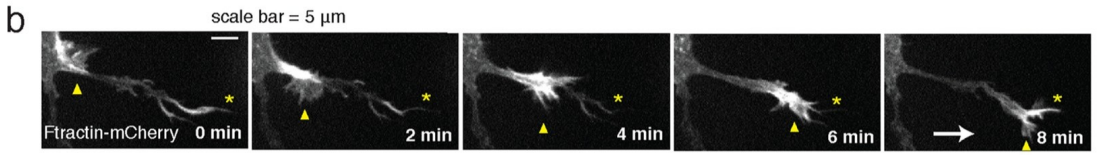
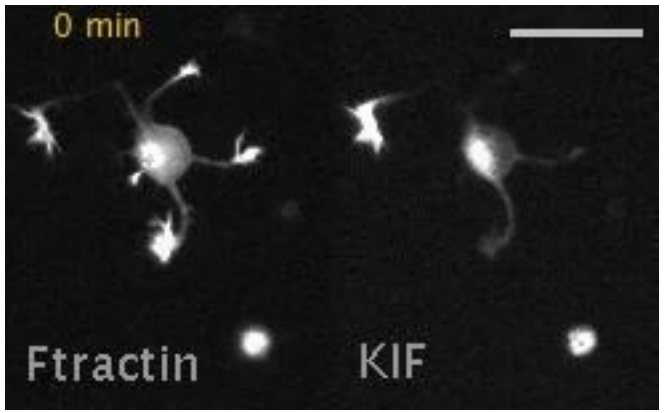
Actin waves accompany bursts of neurite growth

- **Actin waves** are moving regions of **filamentous actin** that travel outward along immature neuronal projections called **neurites**.
  - A **neurite** can later develop into an axon or dendrite; during early development, several neurites can compete to grow.
  - Winans and colleagues tracked fluorescently labeled actin waves in cultured **rat hippocampal neurons**.
  - The waves move from the cell body toward the **growth cone** at the neurite tip.
- As an actin wave passes, the **neurite shaft briefly widens**, creating room for more **microtubules** to grow.
  - **Microtubules** are dynamic internal polymers that help support the process and provide tracks for cargo transport.
  - **Kinesin motor proteins** move cargo along microtubules, so local microtubule growth is linked to increased transport.
- **Neurite growth occurs in bursts** when actin waves arrive near the tip, rather than continuing at one constant rate.
  - The **growth cone** may extend rapidly after a wave arrives, then slow when the wave is gone.
  - Waves enter different branches at different times; this produces **uneven growth** among neurites of the same neuron.
  - During early imaging, approximately **90% of neurite outgrowth** occurred alongside arriving actin waves.



Winans et al. (2016), Fig. 1b. Developing neurites, actin, and intracellular transport.



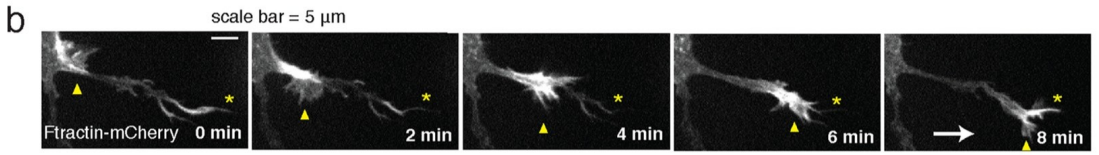
Winans et al. (2016), Video 5. Academic: Neurite growth and transport.

Actin waves accompany bursts of neurite growth

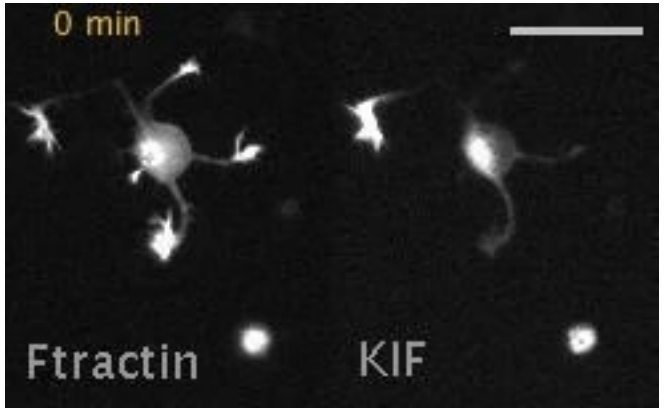
Immature **hippocampal neurons** extend several **neurites**, which are growing projections that can later become axons or dendrites. Within a neurite, **filamentous actin** can reorganize into a moving **actin wave**. Winans and colleagues used fluorescent labeling to track these waves in cultured rat neurons. Individual waves moved outward from the cell body toward the growing tip, while different branches received waves at different times.

As an actin wave travels through a neurite, it briefly **widens the neurite shaft**. The increased space allows additional **microtubules** to polymerize inside the process. These dynamic polymers both support the neurite and provide tracks for **kinesin motor proteins**, which carry cellular cargo toward the tip. Microtubule growth and kinesin-based transport therefore increase in regions reached by the traveling wave.

Neurite extension occurs in bursts associated with an actin wave reaching the **growth cone**. A branch may remain nearly the same length, then extend more rapidly as a wave arrives. Because waves travel into individual neurites at different times, the same developing neuron can have one actively growing branch while another grows more slowly. These changing patterns of transport and extension occur before the neurites acquire their mature identities.



Winans et al. (2016), Fig. 1b. Developing neurites, actin, and intracellular transport.



Winans et al. (2016), Video 5. Academic: Neurite growth and transport.